

Support vector machines – Lesson overview

Exploring support vector machines (SVMs) is essential for **mastering classification tasks**. SVMs provide a robust framework for separating data points into distinct categories, making them **indispensable in various domains**.

In this lesson, we will examine support vector machines and the art of model tuning. We will gain insights into the **working principles of SVMs** and the **role of kernels**. Using both linear and non-linear SVM classifiers in sklearn, we will explore the **significance of hyperplanes** and their **visualisation**. Moreover, we will learn to fine-tune SVM models effectively using GridSearchCV, thereby optimising model performance.

Learning objectives

- Understand the fundamentals of support vector machine models and their operational mechanisms.
- Visualise hyperplanes in SVM classifier models to understand their significance in decision boundary determination.
- Implement SVM classifier models with linear and radial basis function kernels using the sklearn library.
- Tune SVM models using the GridSearchCV function, optimising model parameters for improved performance.



Video



Knowledge questions



Examples



Exercise

